



**S. B. JAIN INSTITUTE OF TECHNOLOGY,
MANAGEMENT & RESEARCH, NAGPUR.**

Session-2025-26 (Even)

TAE-I Project Based Learning

Resume Screening System:

Submitted By

Mohit Mantri (CS23152)

AI Technique(s) (2)	Problem Representation (4)	Tools Explored (2)	Outcome (2)	Total (10)
Sign of Student:			Sign of course In- charge :	



Resume Screening System

How the IT Industry Deals with: Resume Screening System using NLP

1. Industry Problem Context

In the IT industry, organizations receive a large number of resumes for each job opening, making the selection process difficult and time-consuming. Manual screening is inefficient and may lead to errors or bias.

- Large IT companies receive thousands of applications per job
- Manual screening requires significant time and human effort
- Difficulty in matching candidate skills with job requirements
- Chances of inconsistency and bias in decision-making
- Delays in hiring affect overall productivity

Objective:

- Automate the resume screening process
- Identify the most relevant candidates quickly
- Improve accuracy in candidate selection
- Reduce time, cost, and manual workload

2. Feasibility in IT Systems

Resume screening systems are highly feasible due to the availability of digital data and modern processing techniques.

- Resumes are available in structured formats like PDF, DOCX, and TXT
- Text extraction tools can easily retrieve content from documents
- NLP techniques convert unstructured text into usable data
- Systems can process large volumes of resumes simultaneously
- Can be integrated into web applications and HR systems

Key feasibility factors:

- High scalability for handling large datasets
- Fast processing using optimized algorithms
- Easy integration with Applicant Tracking Systems (ATS)
- Supports automation and real-time analysis

Thus, NLP-based systems are practical and widely used in modern recruitment systems.



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3. AI Techniques Used in Industry

- **Natural Language Processing (NLP)**
 - Tokenization: Splitting text into words
 - Lemmatization: Converting words to base form
 - Stop-word removal: Removing unnecessary words
 - Named Entity Recognition (NER): Identifying skills, names, and entities
- **Text Classification**
 - Classifies resumes into job categories
 - Helps filter relevant candidates
- **TF-IDF (Term Frequency – Inverse Document Frequency)**
 - Converts text into numerical feature vectors
 - Assigns importance to keywords
- **Cosine Similarity**
 - Compares resume with job description
 - Generates similarity score (0–100%)
- **Generative AI (LLMs)**
 - Provides detailed feedback on resumes
 - Suggests improvements and missing skills

These techniques help in building an accurate and intelligent resume screening system.

4. Representation/ Modeling in Industry Systems

In this system, both resume and job description are represented in numerical form for comparison.

- Uses **Vector Space Model (VSM)**
- Converts text into feature vectors using TF-IDF

Processing Steps:

- Input: Resume and Job Description
- Text Extraction from documents
- Text Preprocessing using NLP
- Conversion into TF-IDF vectors
- Similarity calculation using cosine similarity

Advantages of this model:

- Fast comparison between documents
- Works efficiently with large datasets
- Suitable for real-time applications



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5. Tools & Technologies in Industry

- **Programming Language:**
 - Python (used for backend and AI processing)
- **Backend Framework:**
 - Flask (handles server operations and APIs)
- **Frontend Technologies:**
 - HTML (structure)
 - CSS (design and layout)
 - JavaScript (interaction and dynamic content)
- **Natural Language Processing:**
 - spaCy (text processing and analysis)
- **Machine Learning:**
 - scikit-learn
 - TfidfVectorizer (feature extraction)
 - cosine_similarity (similarity measurement)
- **Numerical Computation:**
 - NumPy (efficient calculations)
- **File Handling:**
 - PyPDF2 (PDF parsing)
 - python-docx (DOCX parsing)
- **AI Integration:**
 - Groq API (Llama 3 model for feedback generation)
- **Deployment:**
 - Virtual Environment (venv)
 - Can be deployed on cloud platforms like AWS or GCP

This technology stack ensures efficiency, scalability, and real-time performance.

6. Industry-Level Outcome

- Automated resume screening in a few seconds
- Accurate generation of skill match score (%)
- Identification of matching and missing skills
- Reduction in manual effort and human error
- Faster hiring decisions and improved efficiency
- Consistent and unbiased evaluation of candidates

Performance Metrics:

- Match accuracy (%)
- Processing time per resume
- Candidate ranking quality
- Skill extraction accuracy

7. Real-World Industry Example/use

- HR platforms use it for automated screening
- IT companies use it for bulk hiring processes
- Job portals use it to match candidates with jobs
- Recruitment agencies use it to filter applicants
- ATS systems use it to manage hiring workflows

Example (ResumeIQ Project):

- User uploads resume and job description
- System extracts and processes text
- Applies NLP and TF-IDF techniques
- Computes similarity score
- Displays matched and missing skills
- Generates AI-based feedback



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8. Real-World Industry Example/use

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Final Result:

An intelligent system that provides fast, accurate, and unbiased resume screening to support better hiring decisions.

The screenshot shows the homepage of the 'AI Screen' Resume Screening System. The header includes the 'AI Screen' logo, navigation links for Features, Workflow, Demo, Pricing, and a 'Get Started' button. Below the header, there's a section titled 'Resume Screening System with Vector Search.' with a subtitle 'Automate resume filtering using NLP-based text classification and Feature Vectors. Get precise skill match scores in seconds.' To the right of the text is an image of a laptop displaying a dashboard with charts and a 'Skill Match: 98%' indicator. At the bottom, there are two buttons: 'Start Screening →' and 'View Architecture'.

Powerful Features for Modern Teams

Everything you need to find the perfect candidate using advanced NLP techniques.



Feature Vector Matching

Utilizes TF-IDF vectorization and cosine similarity to mathematically calculate the precise alignment between resumes and job descriptions.



NLP-Based Classification

Go beyond keywords. Our Python-based NLP engine classifies text to understand context, seniority, and domain expertise.



Skill Match Scores

Get a quantifiable match score (0-100%) for every candidate, backed by vector distance metrics and semantic analysis.



Bias Reduction

Ensure fair hiring with algorithmic screening that focuses purely on feature vectors and skill sets, removing unconscious bias.



High-Volume Processing

Process thousands of applications in minutes using optimized Python libraries like Scikit-learn and NumPy.



Secure & Compliant

Enterprise-grade data protection ensures all candidate information is kept safe and private.

How It Works

A simple, streamlined process to elevate your recruitment game.



Upload Resumes

Deep-load bulk PDFs or Docx files directly into our secure processing engine.



NLP Analysis

Our Python-based NLP engine extracts skills, experience, and intent from every application.



Scoring & Ranking

Candidates are instantly ranked based on job-specific relevance scores.



Hire the Best

Export shortlist reports and start interviewing the top 1% immediately.

Scientific Foundation: System 37

Our platform is built upon robust NLP principles and mathematical feature vectorization to solve the core problem of large-scale resume filtering.

99.8%

SIMILARITY ACCURACY

< 0.5s

PROCESSING TIME



PROBLEM

Resume filtering



FEASIBILITY STUDY

NLP-based



AI TECHNIQUE

Text classification



REPRESENTATION

Feature vectors (TF-IDF)



TOOLS

Python (Scikit-learn, NumPy)



OUTCOME

Skill match score (0-100%)

The Smarter Way to Shortlist

Experience the power of semantic search and intelligent ranking. Our dashboard provides a clear overview of candidate suitability based on your specific requirements.

- Instant semantic matching
- Advanced filtering & sorting
- Direct collaboration tools
- Export-ready reports

Explore Dashboard

Search Candidates...		
J	John Doe Senior Frontend Engineer	94% Match High
J	Jane Smith Full Stack Developer	88% Match High
M	Mike Ross Product Manager	42% Match Low

AI Screen.

Features Workflow Demo Pricing

Get Started

AI-POWERED FEATURE VECTOR ANALYSIS

Resume Screening System

Upload resumes to automatically screen and rank candidates based on skill match scores using NLP and vector similarity.

JOB DESCRIPTION

Frontend React Full Stack Node Data Scientist DevOps Engineer Security Analyst

Paste your job description here or upload template above...

RESUMES (PDF/DOCX)

1 SELECTED

Click to Upload Resumes
Drag & drop multiple files

No resumes uploaded yet

Calculate Match Scores

AI Screen.

Features Workflow Demo Pricing

Get Started

AI-POWERED FEATURE VECTOR ANALYSIS

Resume Screening System

Upload resumes to automatically screen and rank candidates based on skill match scores using NLP and vector similarity.

JOB DESCRIPTION

Frontend React Full Stack Node Data Scientist DevOps Engineer Security Analyst

Full Stack Developer
Skills: Node.js, Express, PostgreSQL, React, AWS, Docker.
Experience: 3+ years in full-lifecycle web development.

RESUMES (PDF/DOCX)

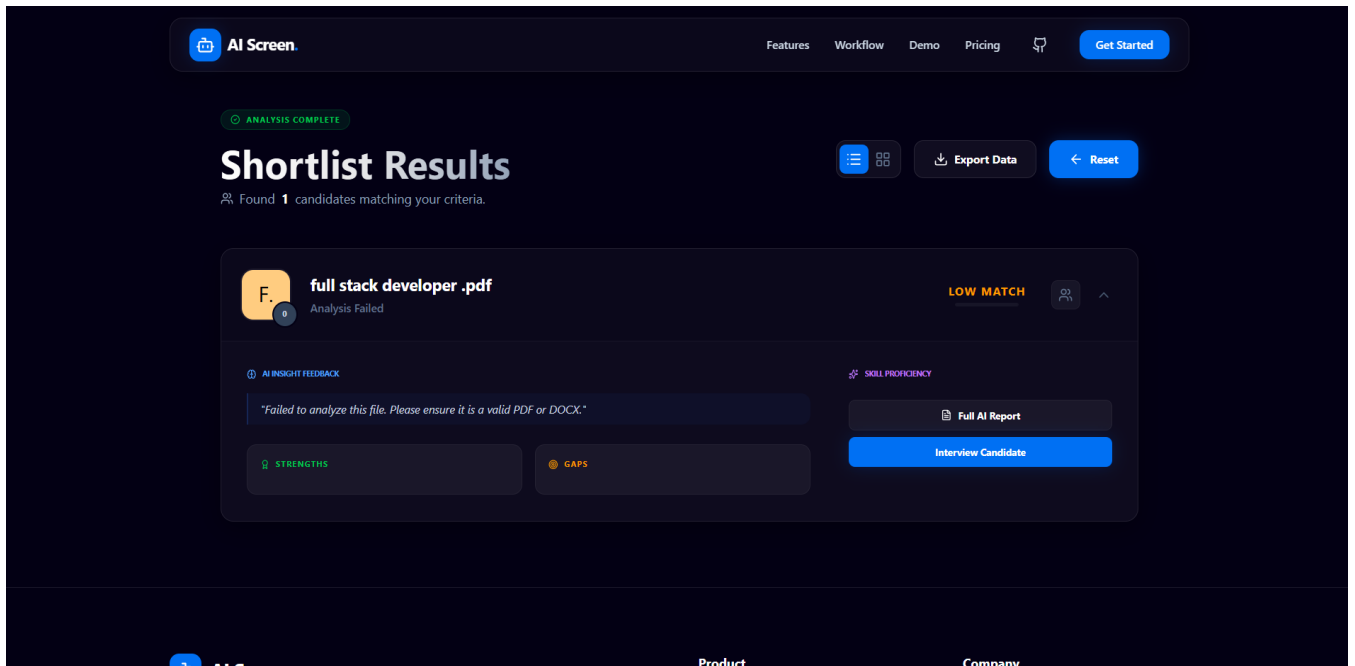
1 SELECTED

ANALYSIS PROGRESS 5%

Uploading & Analyzing full stack developer .pdf...

Processing...

Secure Analysis: Files are processed securely via NLP.



9. References:

- https://scikit-learn.org/stable/modules/feature_extraction.html
- <https://spacy.io/usage>
- <https://docs.python.org/3/>
- <https://pypi.org/project/PyPDF2/>
- <https://pypi.org/project/python-docx/>
- <https://console.groq.com/docs>